MILITARY SPECIFICATION

ADAPTION KIT, ATOMIC WARHEAD TO GUIDED MISSILE: M238

1. SCOPE

1.1 This specification covers special requirements not included on the drawings and provides quality assurance provisions for the fabrication of parts, assembling and the preparation for delivery for one type of assembly used to house a warhead, provide fuzing and arming signals, and designated as Adaption Kit, Atomic Warhead to Guided Missile: M238, which is used in the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-48524	- "A" Section Assembly.
MIL-A-45866	- AK Subassembly.
MIL-C-48526	- Cable Assemblies.
MIL-S-48525	- Container, Shipping and Storage Warhead Section, M511.
MIL-F-60972	- Fuzes, Altitude, M1140.
MIL-F-48306	- Impact Fuze, M1139.
MIL-C-48523	- Skins and Structures - Ballistic Case Sections "C" and "D".
MIL-T-48527	- Test Set, T4186.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.

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DRAWINGS

ORDNANCE CORPS

8880000	- Adaption Kit, Atomic Warhead to Guided Missile: M238.
9216660	- Shipping and Storage Container for Lance Warhead Sections and Adaption Kits, Packaging and Marking.
9237200	- Painting and Marking, AK, Atomic (M238).
9237212 (sheet 1 only)	- Simulated Warhead Section.
9237577	- Ballast Assembly.
EL-8880000	- Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The adaption kit shall comply with all requirements specified on the drawing (dwg) 8880000, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.

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c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).
e. The security classification of the number of serial codes requested only if this number reveals classified production quantities.

3.5 Components. For the components to be used in the adaption kit, objective evidence indicating that the following components conform to their respective specification, listed below, shall be made available to the Government representative:

a. T4186 Test Set MIL-T-48527.
b. "A" Section Assembly MIL-C-48524.
c. Skins and Structures - Ballistic Case Sections "C" and "D" MIL-C-48523.
d. Fuzes, Altitude, M1140 - MIL-F-60972.
e. AK Subassembly - MIL-A-45866.
f. Cable Assemblies - MIL-C-48526.
g. Impact Fuze, M1139 - MIL-F-48306.
h. Container, Shipping and Storage, Warhead Section, M511 - MIL-S-48525.

3.6 Continuity loop. The continuity loop of the Assembled adaption kit shall be verified when tested in accordance with 4.4.1. CAUTION: USE EXTREME CARE IN TESTING TO PRECLUDE ANY DAMAGE TO SEQUENTIAL TIMER.

3.7 Circuit resistance. The resistance of each circuit shall not exceed the maximum value listed in Table I or as shown otherwise when tested as specified in 4.4.2. CAUTION: USE EXTREME CARE IN TESTING TO PRECLUDE ANY DAMAGE TO SEQUENTIAL TIMER.

Table I

Terminals From	Terminals To	Resistance Ohms (max)	Option Switch Setting*	Test Code No. (see 3.14)
1J1-A	1J1-U	0.3	A-1	060001
1J1-B	1J13-47	0.3	A-1	060002
1J1-C	1J1-V	0.3	A-1	060003
1J1-C	1J1-S	0.3	A-1	060004
1J1-C	1J1-d	0.3	A-1	060005
1J1-D	1J1-$\bar{E}$	0.3	A-1	060006
1J1-F	1J1-Z	0.3	A-1	060007
1J1-F	1J1-h	0.3	A-2	060008
1J1-F	1J1-$\bar{h}$	0.3	A-5	060009
1J1-F	1J1-$\bar{h}$	0.3	A-3	060010
1J1-F	1J1-$\bar{h}$	0.3	A-6	060011
1J1-F	1J1-$\bar{R}$	0.3	A-3	060012
1J1-F	1J1-R	0.3	A-6	060013

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Table I (con't)

1J1-F	1J1-J	0.3	B-2	060014
1J1-F	1J1-J	0.3	B-5	060015
1J1-F	1J1-f	0.3	C-2	060016
1J1-F	1J1-f̄	0.3	C-5	060017
1J1-F	1J1-N̄	0.3	C-3	060018
1J1-F	1J1-N	0.3	C-6	060019
1J1-G	1J1-Y	0.3	A-1	060020
1J1-G	1J1-j	0.3	A-2	060021
1J1-G	1J1-j̄	0.3	A-5	060022
1J1-G	1J1-j	0.3	B-3	060023
1J1-G	1J1-j	0.3	B-6	060024
1J1-G	1J1-P̄	0.3	A-3	060025
1J1-G	1J1-P	0.3	A-6	060026
1J1-G	1J1-a	0.3	B-2	060027
1J1-G	1J1-ā	0.3	B-5	060028
1J1-G	1J1-q	0.3	C-2	060029
1J1-G	1J1-q̄	0.3	C-5	060030
1J1-G	1J1-c̄	0.3	C-3	060031
1J1-G	1J1-c	0.3	C-6	060032
1J1-T	1J1-e	0.3	A-1	060033
1J2-A	1J2-J̄	0.3	A-1	060034
1J2-B	1J13-42	0.3	A-1	060035
1J2-B	1J1-H	0.3	A-1	060036
1J2-C	1J2-R	0.3	A-1	060037
1J2-D	1J2-X	0.3	A-1	060038
1J2-D	1J13-50	0.3	A-1	060039
1J2-E	1J2-Y	0.3	A-1	060040
1J2-E	1J13-6	0.3	A-1	060041
1J2-F	1J2-W	0.3	A-1	060042
1J2-G	1J2-M	0.3	A-1	060043
1J2-H	1J13-30	0.3	A-1	060044
1J2-K	1J2-U	0.3	A-1	060045
1J2-L	1J2-V	0.3	A-1	060046
1J2-N	1J2-P	0.3	A-1	060047
1J2-S	1J13-46	0.3	A-1	060048
1J2-a	1J13-43	0.3	A-1	060049
1J2-L̄	1J1-C	0.3	A-1	060051
1J12-B	1J12-V	0.3	A-1	060052
1J12-D	1J13-16	0.3	A-1	060053
1J12-E	1J13-17	0.3	A-1	060054
1J12-F	1J12-G	0.3	A-1	060055
1J12-F	1J12-H	0.3	A-1	060056
1J12-F	1J12-J	0.3	A-1	060057
1J12-F	1J12-X	0.3	A-1	060058
1J12-F	1J12-Y	0.3	A-1	060059
1J12-F	1J12-Z	0.3	A-1	060060
1J12-F	1J12-f	0.3	A-1	060061
1J12-R	1J12-e	0.3	A-1	060062
1J12-q	1J13-66	0.3	A-1	060063
1J12-h	1J13-67	0.3	A-1	060064

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Table I (con't)

1J13-28	1J13-29	0.3	A-1	060065
1J13-28	1J13-31	0.3	A-1	060066
1J13-61	1J13-64	0.3	A-1	060067
1J13-33	1J13-34	0.3	A-1	060068
1J13-36	1J13-34	0.3	A-1	060069
1J13-36	1J13-37	0.3	A-1	060070
1J13-45	1J13-49	0.3	A-1	060071
1J13-45	1J13-52	0.3	A-1	060072
1J13-53	1J13-52	0.3	A-1	060073
1J13-53	1J13-54	0.3	A-1	060074
1J13-55	1J13-54	0.3	A-1	060075
1J13-44	1J13-49	0.3	A-1	060076
1J13-38	1J13-40	0.3	A-1	060077
1J13-35	1J13-39	0.3	A-1	060078
1J13-38	1J13-37	0.3	A-1	060079
1J13-30	1J13-29	0.3	A-1	060080
1J13-42	1J1-H	0.3	A-1	060081
1J13-44	1J13-40	0.3	A-1	060082
1J2-D	1J2-K	4-7	A-1	060083
1J2-D(+)	1J2-a	50-70	A-1	060084
1J2-D(-)	1J2-$\bar{a}$	180-220	A-1	060085
1J2-K(+)	1J2-$\bar{S}$	50-70	A-1	060086
1J2-K(-)	1J2-S	180-220	A-1	060087
1J2-S(+)	1J2-a	250-350	A-1	060088
1J2-S(-)	1J2-$\bar{a}$	250-350	A-1	060089
1J12-F	1J12-B	5-7	B-1	060090
1J12-F	1J12-B	5-7	C-1	060091
1J13-6	1J12-27	24-32	A-1	060092
1J13-8(+)	1J13-10	15-30	A-1	060095
1J13-8(-)	1J13-10	15-30	A-1	060096
1J13-8(+)	1J13-9	15-30	A-1	060097
1J13-8(-)	1J13-9	15-30	A-1	060098
1J13-3(+)	1J13-24	10-20	A-1	060099
1J13-12	1J13-24	270-330	A-1	060100
1J13-11	1J13-24	270-330	A-1	060101
1J13-13(-)	1J13-25	70-90	A-1	060102

*The option switch setting is as follows:

A B C Switch	Option Switch
A	1 = S
	2 = HG
B	3 = LG
	4 = G
C	5 = H
	6 = L

(+) (-) Polarity of Measuring Device.

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3.8 Insulation resistance. The insulation resistance between each circuit and all other circuits, as indicated in Table II, shall not be less than 100 megohms when tested as specified in 4.4.3. CAUTION: USE EXTREME CARE IN TESTING TO PRECLUDE ANY DAMAGE TO SEQUENTIAL TIMER.

Table II

From	To	From	To
1J2-A	All others & shells	1J1-F	All others & shells
1J2-H	"	1J1-G	"
1J2-C	"	1J1-H	"
		1J1-J	"
1J2-F	"	1J1-K	"
1J2-G	"		
1J2-N	"		
1J2-b	"	1J1-L	"
1J13-22	"	1J1-N	"
1J13-16	"	1J1-P	"
1J13-17	"	1J1-R	"
1J1-B	"	1J1-f	"
1J1-C	"	1J1-g	"
1J1-c	"	1J1-h	"
		1J1-J	"
		1J1-a	"
		*1J13-23	"
			"

The following list of unused pins may be connected together and tested as one terminal:

1J2-T, Z
1J12-A, C, S, T, M, P, U, W, a, b, d, j,
1J13-2, 4, 19*, 20*, 21*, 26, 41

*Used on ET/ST Units only.
**The Option switch must be in the (S) safe position and the ABC Switch must be in the "A" position for this test.
CAUTION: USE EXTREME CARE IN TESTING TO PRECLUDE ANY DAMAGE TO SEQUENTIAL TIMER.

3.9 Environmental testing. The adaption kit shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The adaption kit shall also comply with 3.6 thru 3.8 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

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3.9.1 Low temperature. The assembled adaption kit with 9237577 Ballast shall withstand a temperature of -40° ± 5°F for a minimum of four (4) hours after stabilization (see 6.4) without evidence of cracking, splitting, separation of parts, or any other damage and shall comply with the requirements of 3.6, 3.7 and 3.8. Upon completion of this test, the adaption kit shall be allowed to return to ambient temperature (80°F ± 20°F) without any forced heating and comply with post test requirements as specified in 3.6, 3.7 and 3.8. Testing shall be as specified in 4.4.4.1.

3.9.2 High temperature. The assembled adaption kit with 9237577 Ballast shall withstand a temperature of 140° ± 5°F for a minimum of four (4) hours after stabilization (see 6.4), without evidence of cracking, splitting, separation of parts, or any other damage. After stabilization, the adaption kit shall comply with the requirements of 3.6, 3.7 and 3.8 when tested in accordance with paragraph 4.4.4.2. Upon completion of this test, the adaption kit shall be allowed to return to ambient temperature (80° ± 20°F) without any forced cooling and comply with post test requirements as specified in 3.6, 3.7 and 3.8.

3.9.3 Shock. The adaption kit mated to a Ballast Assembly (see dwg 9237577) and installed in its container M511 container shall withstand the following tests. The mating of the ballast assembly shall be as shown on dwg 9237212 sheet 1 only:

Test	Para. (See MIL-S-48525)
a. Cornerwise Drop	4.4.4.1.1
b. Edgewise Drop	4.4.4.1.2
c. Impact	4.4.4.1.3
d. Roll Over Impact	4.4.4.1.4

Testing is to be conducted in accordance with 4.4.4.3. Upon completion of the tests, the adaption kit shall display no evidence of damage. It shall meet the requirements of 3.6, 3.7 and 3.8.

3.9.4 Vibration.

3.9.4.1 Transportation vibration (at 80°F + 20°F). The adaption kit mated to a Ballast Assembly (dwg 9237577) in accordance with dwg 9237212 sheet 1 only shall withstand sinusoidal vibration conditions specified in Table III in the longitudinal axis only when tested as specified in 4.4.4.4.1. Frequency shall be cycled at a logarithmic sweep rate between 5-500-5 Hertz (Hz) in 15 minutes.

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Table III

Resonant Search Vibration

Frequency (Hz)	Gravity Units (g's) Double Amplitude (da-in)	Time-Minutes and Sweep Rate
10-14	0.1 da ± .01	At ambient temperature, 1 frequency sweep from 10-500-10 at 0.745 octave/min.
14-500	0.1 g's ± 0.1	

Sinusoidal vibration

Frequency (Hz)	Gravity Units (g's) Double Amplitude (da-in)	Time-Minutes and Sweep Rate
5-17	0.2" da ± 0.02	Frequency sweep at 0.875 octave/min at ambient temperature, 3 frequency sweeps from 5-500-5 (15 min each sweep) for a total of 45 min.
17-130	3.0 g's ± 0.3	
130-500	5.0 g's ± 0.5	

Upon completion of the tests, the adaption kit shall display no evidence of damage. It shall meet the requirements of 3.6, 3.7 and 3.8.

3.10 Quick access door operation. The quick access door assembly shall be capable of being operated without undue effort when tested in accordance with 4.4.5.

3.11. Parameter setting door operation. The parameter setting door assembly shall be capable of being operated without undue effort when tested in accordance with 4.4.6.

3.12 Weight and center of gravity. The weight and center of gravity shall comply with the requirement as shown on dwg 8880000 when tested as specified in 4.4.8.

3.13 Assembly. There shall be no undue effort required or other tools used than those specified for disassembly and re-assembly of the adaption kit in accordance with 4.4.7. There shall be no damage to any component or ablative material.

3.14 Test identification number. The contractor shall supply feedback data on punch cards using six digit number test code No. designated in the applicable paragraph in Section 3. The feedback data shall be prepared and mailed in accordance with QAI-PA-1.

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3.15 Capacitance. Prior to mating of the 1P16 and 1P17 connectors to the AK Subassembly M239, the capacitance between the center conductors and the connector shells as listed in Table IV shall be as specified when tested in accordance with 4.4.9.

Table IV

*Capacitance - 575 ± 75 picofarads (pf)
at 1000 Hz ± 50 Hz

From Shell	To Center Conductor	Test Code No. (see 3.14)
1P16	1P16	019001
1P17	1P17	019002

3.16 Workmanship.

3.16.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.16.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.16.3 Burrs. Parts shall be free from burrs.

3.16.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.16.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.16.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8880000) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table V. Tests shall be conducted in order specified in Table V.

Table V
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Continuity loop (see 3.6)		X
3. Circuit resistance (see 3.7)		X
4. Insulation resistance (see 3.7)		X
5. Low temperature (see 3.9.1)		X
6. High temperature (see 3.9.2)		X
7. Shock (see 3.9.3)		X
8. Vibration (see 3.9.4)	X	
9. Assembly (see 3.13)		X
10. Capacitance (see 3.15)		X
11. Quick access door operation (see 3.10)		X
12. Parameter setting door operation (see 3.11)		X
13. Weight and center of gravity (see 3.12)		X
14. Components (see 3.5)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturor in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

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4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VI and Table VII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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112. Incorrect torque for station 100 connector.....Gage
113. Incorrect torque for aft cover.....Gage
114. Yellow ribbon and connector assembly loose or missing (2 places).....Visual
115. Lettering on yellow ribbon incorrect, illegible or missing (both sides) (2 places).....Visual
116. Markings missing, illegible, or incorrect.....Visual
117. Incorrect forward and aft facing steps.....Gage
118. Incorrect torque on maintenance door mounting bolts (26).....Gage

Minor:

201. Evidence of poor workmanship (see 3.16).....Visual

4.3.2.1.2 Packing for adaption kit assembly in M511 shipping and storage container prior to closing cover (see dwg 9216660).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Adaption kit ablative material and protective finish damaged (see dwg 8880000).....Visual
102. Adaption kit component and hardware loose or missing, (see dwg 8880000).....Visual
103. Adaption kit guide pins not properly engaged in Station 100 support plate....Visual
104. Incorrect torque at Station 100 swing bolts at support plate (4 places).....Gage
105. Lower strap liner for lower strap cushion missing or incorrect, permitting strap cushion to come in contact with AK.....Visual
106. Upper strap liner missing allowing upper strap cushion to come in contact with AK.. .....Visual
107. Incorrect torque for securing upper strap assembly over AK.....Gage
108. Protective cover assembly torque incorrect.....Gage
109. A2P2 Connector of W58U Cable not mated to A2J2 Connector on Adaption Kit and PAL access cover not mated to storage connector on support plate.....Manual/Visual

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110. Incorrect torque for humidity indicator.....Gage
111. Incorrect torque for valves (2).....Gage
112. Incorrect torque for W58U cable.....Gage
113. Dessicant insufficient.....Visual
114. Tee handle wrench missing.....Visual
115. Gasket nicked, cut, or damaged to the extent of rendering it unusable.....Visual
116. Marked mailing bags with con ents missing or improper in storage box.....Visual
117. Hardware in bottom assembly container incorrectly torque (see table V on dwg 9216660).....Gage
118. Any component part missing, damaged, improperly assembled, or unserviceable.....Visual

Minor:
201. Evidence of poor workmanship (see 3.16).....Visual

4.3.2.1.3 Packing for adaption kit assembly in M511 shipping and storage container after closing of cover.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Flange bolts torque incorrect (22).....	Gage
102.	All external hardware on exterior of container torque incorrect (see table on dwg 9216660).....	Gage
103.	EML card, Item Data Card, and two wrenches incorrect or missing in Records Container Assembly.....	Visual
104.	Lead wire seals missing (9) required...	Visual
105.	Records container cover loose, missing, unserviceable, gasket missing unserviceable.....	Visual
106.	H4244 wrench(es) unserviceable, missing, mutilated.....	Manual/Visual
107.	Markings illegible or do not agree with dwg 9216660 and EML.....	Visual
108.	Structural damage which will preclude use of container for intended purpose.....	Visual
109.	Paint peeling, scratched, inadequate coverage, wrong color.....	Visual
110.	Hardware missing, unserviceable, loose.....	Manual/Visual
111.	Pressure equalizing valves loose, unserviceable.....	Manual/Visual

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112. Humidity indicator card reads 30 percent or more.....Visual
113. Humidity indicator assembly loose, unserviceable.....Manual/Visual
114. Quick access cover loose, damaged, unserviceable.....Manual/Visual
115. Presence of corrosion, fungus and other foreign material.....Visual
116. External PAL connector cover missing or unserviceable.....Visual
117. Lift bars and handles cracked, broken or loose.....Visual
118. Lift handle clips unserviceable or missing.....Visual
119. Fork lift bars missing, loose or damaged.....Visual
120. Stacking bolt holes, skid guides on cover and grooves on skids missing or incorrectly located.....Visual
121. Desiccant assess cover loose, unserviceable.....Manual/Visual
122. Wooden runner (skid) unserviceable, splits more than 3" long regardless of depth.....Gage/Visual
123. Identification plate missing, FSN Ser No., Contract No., nomenclature, date of manufacturing, missing or illegible.....Visual

Minor:
201. Evidence of poor workmanship (see 3.16).....Visual

4.3.3 Testing. All tests shall be performed. Tests need not performed in the order specified below:

Test	Test classification
a. Continuity loop	Major
b. Circuit resistance	Major
c. Insulation resistance	Major
d. Quick access door operation	Major
e. Parameter setting door operation	Major
f. Weight and center of gravity	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Continuity loop, circuit resistance, insulation resistance, quick access door operation, parameter setting door operation, and weight and center of gravity. Each adaption kit shall be tested in accordance with 4.4.1 through 4.4.3 and 4.4.5, 4.4.6 and 4.4.8.

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If any item fails to comply with 3.6 through 3.8 and 3.10 through 3.12, the item shall be rejected and removed from the lot.

Note When the same contractor manufacturers both the M239 and M238 at the same facility, the circuit resistance (3.7) and insulation resistance (3.8) tests will not be required.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8880000). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Continuity loop. The continuity loop of the assembled adaption kit shall be verified by utilizing the T4186 test set and the applicable test procedures.

4.4.2 Circuit resistance. By removing the end cover at Station 100, the circuit resistance as specified in 3.7 shall be measured utilizing a suitable circuit resistance tester with current limited to 10 milliamperes (ma) ± 0.10 ma

4.4.3 Insulation resistance. An insulation resistance tester, capable of supplying 500 volts direct current (vdc) ± 50 vdc (applied for 5 seconds min and 60 seconds max) shall be used to measure the insulation resistance. The current shall be limited to 5 ma. The measured resistance shall be as specified in 3.8. Remove end cover at Station 100.

4.4.4.1 Low temperature. The adaption kit shall be placed in a suitable test chamber. The temperature within the chamber shall be lowered to -40°F ± 5°F and held at this temperature for a minimum of four hours after the temperature of the adaption kit has been stabilized (see 6.4).

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After stabilization, the test unit shall show no evidence of damage. It shall meet the requirements of 3.9.1 while at this temperature. Upon completion of the test, the adaption kit shall be capable of meeting the requirements of 3.9.1 at ambient temperature.

4.4.4.2 High temperature. The adaption kit shall be placed in a suitable test chamber. The temperature within the chamber shall be raised to 140°F ± 5°F and held at this temperature for minimum of four hours after the temperature of the adaption kit has stabilized (see 6.4). After stabilization at the required high temperature, the unit shall show no evidence of damage. It shall meet the requirements of 3.9.2 while at this temperature. Upon completion of the tests, the adaption kit shall be capable of meeting the same requirements of 3.9.2 at ambient temperature.

4.4.4.3 Shock. The adaption kit shall be tested in accordance with 4.4.4.1 of MIL-S-48525.

4.4.4.4 Vibration.

4.4.4.4.1 Transportation vibration (at 80°F + 20°F). The adaption kit mated to a Ballast Assembly (see dwg 9237577) in accordance with dwg 9237212 Sheet 1 only and shall be mounted securely to a suitable fixture and vibrated at the levels and conditions specified in 3.9.4.1. Control of the "g" or double amplitude (da) levels shall be held to the total complex vibration signal. No tracking filter is required.

4.4.5 Quick access door operation. The quick access door assembly shall be capable of being operated and removed by pushing the button per marking and rotating the handle one quarter turn in the counterclockwise direction.

4.4.6 Parameter setting door operation. The parameter setting door assembly shall be capable of being operated and removed by rotating the center socket one quarter turn in the direction of the marking arrow.

4.4.7 Assembly. The adaption kit shall be disassembled and visually inspected for any damage. Upon satisfactory disassembly, inspection and recording of any damage, the adaption kit shall be re-assembled in accordance with dwg 8880000.

4.4.8. Weight and center of gravity. The weight and center of gravity (cg) of each AK M238 Assembly shall be measured using suitable equipment and fixtures. The weight and cg shall be recorded for each AK assembly. Weight and cg is specified on 8880000.

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4.4.9 Capacitance. The capacitance between center pin and connector shell shall be measured using a Government approved electrical instrument with a suitable coaxial connector and capable of measuring picofards. The capacitance shall be within the values specified in 3.15.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The adaption kit shall be packaged in accordance with dwg 9216660 using the specified moisture barrier.

5.1.2 Level C. The adaption kit shall be packaged in accordance with dwg 9216660 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The adaption kit shall be packed in accordance with dwg 9216660.

5.3 Marking. Marking shall be as specified in accordance with dwg 9237200.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 3.14 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-1197A dated 21 November 1972, and amendment 8 dated 16 April 1975.

Custodian:
Army - PA

Preparing activity:
Army - PA

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